

MAT 426: Topics in Linear Algebra

Exam 1

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- *You are NOT ALLOWED to use your book, notes and homework*
- *You are NOT ALLOWED use internet/web resources.*
- *You are NOT ALLOWED to consult with others.*
- *You are allowed to use a scientific, non graphing calculator.*

Honesty Pledge

I affirm that the work I am submitting upholds the Centre College standard for academic honesty, that it is solely my own work, and that I have neither given nor received any unauthorized assistance from other people or resources.

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1. (10 points) Let

$$A = \begin{bmatrix} 2 & 1 & 1 & 3 \\ 4 & 3 & 3 & 7 \\ 8 & 7 & 9 & 19 \\ 6 & 7 & 9 & 20 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 5 \\ 4 \end{bmatrix}$$

(a) Find the LU factorization $A = LU$ where L is unit lower triangular.

(b) Use the factorization to solve the system $A\mathbf{x} = \mathbf{b}$.

2. (8 points) Answer the following by circling either TRUE or FALSE.

- | | | |
|------|-------|---|
| True | False | If a matrix A has an LU factorization, then the factorization is unique. |
| True | False | Finding the LU factorization of an $n \times n$ matrix A requires $\frac{2}{3}n^3 + \mathcal{O}(n^2)$ flops. |
| True | False | Gaussian elimination may fail if a pivot is zero unless row swaps are allowed. |
| True | False | When implemented in a computer, Gaussian elimination without pivoting will always produce matrices LU with $A = LU$ as long as a zero pivot is not encountered. |
| True | False | If $\mathbf{u}$ and $\mathbf{v}$ are orthogonal vectors, then $\ \mathbf{u} + \mathbf{v}\ ^2 = \ \mathbf{u}\ ^2 + \ \mathbf{v}\ ^2$. |
| True | False | If U is an orthogonal matrix, then $U^{-1} = U^T$. |
| True | False | If $\mathbf{u} \cdot \mathbf{v} = 0$, then $\mathbf{u}$ and $\mathbf{v}$ must both be nonzero. |
| True | False | The projection of a vector onto another vector is always orthogonal to that vector. |

3. (3 points) Complete the following statements.

- (a) The product of a permutation matrix with itself produces an _____ matrix.
- (b) The product of two permutation matrices is a _____ matrix.
- (c) Multiplying a elementary matrix M on both the left and right by a permutation matrix P produces a _____ matrix.

4. (6 points) Gaussian elimination with partial pivoting directly yields the factorization $MA = U$ where M is a permuted elementary matrix and U is an upper triangular matrix. For a 3×3 matrix A , show how M is formed and how the $PA = LU$ factorization is extracted.

5. (12 points) Suppose A , L and U are $n \times n$ matrices.

(a) Show that solving an upper triangular system $U\mathbf{x} = \mathbf{y}$ requires n^2 floating point operations (flops).

(b) How many flops are required to solve a unit lower triangular system $L\mathbf{y} = \mathbf{b}$?

(c) Performing row operations to row reduce $[A|I]$ to $[I|A^{-1}]$ is equivalent to solving n linear systems $A\mathbf{x} = \mathbf{e}_i$, where $\mathbf{e}_i$ is the i th column of the identity matrix I . How many floating point operations are needed to solve these n linear systems using the LU decomposition of A ?

(d) How many floating point operations are required to compute A^{-1} ?

(e) How many floating point operations are needed to multiply $A^{-1}\mathbf{b}$?

(f) Show that $\frac{8}{3}n^3 + \mathcal{O}(n^2)$ floating point operations are needed to solve $A\mathbf{x} = \mathbf{b}$ by finding the inverse of A using the method outlined above.

6. (12 points) Let

$$A = \begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & -1 & 2 & 1 \end{bmatrix}$$

- (a) Use Gaussian elimination with partial pivoting to find the $PA = LU$ decomposition of A . Clearly state the matrices P , L , and U .

- (b) Suppose GEPP is implemented in a computer algorithm. State the matrix A and vector $\mathbf{r}$ that would be returned by this algorithm.

7. (4 points) Define the following.

(a) The inner product of two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is

(b) Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are orthogonal if and only if _____.

8. (11 points) Let $\mathbf{u} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 0 \\ -5 \\ 4 \end{bmatrix}$. Compute the following:

(a) $\mathbf{u} \cdot \mathbf{v}$

(b) $\|\mathbf{u} - \mathbf{v}\|$

(c) Write $\mathbf{v}$ as the sum of a vector in $\text{Span}\{\mathbf{u}\}$ and a vector orthogonal to $\mathbf{u}$

9. (8 points) Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. The Cauchy-Schwarz inequality states that $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \cdot \|\mathbf{v}\|$. Use the Cauchy-Schwarz inequality to prove that $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

10. (12 points) Let $S = \left\{ \mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_4 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ -2 \end{bmatrix} \right\}$ and let $\mathbf{y} = \begin{bmatrix} 3 \\ 2 \\ 1 \\ 4 \end{bmatrix}$.

(a) Show that S is an orthogonal set.

(b) Explain why S forms an orthogonal basis for $\mathbb{R}^4$.

(c) Write $\mathbf{y}$ as a linear combination of the vectors in S .

(d) Is S an orthonormal set? If not, find an orthonormal set S' such that $\text{Span}(S') = \text{Span}(S)$.

11. (8 points) Prove that an orthogonal set of nonzero vectors is linearly independent.

12. (6 points) Consider the matrix $A = \begin{bmatrix} \frac{1}{2} & \frac{1}{\sqrt{3}} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2\sqrt{3}} & -\frac{\sqrt{3}}{4} \\ \frac{1}{2} & -\frac{1}{2\sqrt{3}} & -\frac{1}{4} \\ \frac{1}{2} & -\frac{1}{\sqrt{3}} & \frac{\sqrt{3}}{4} \end{bmatrix}$.

(a) Determine whether or not A is an orthonormal matrix.

(b) Is A an orthogonal matrix? Explain.